CY4001: Industrial Chemistry

Unit 4:Polymer Chemistry
(Lecture 2 – Macromolecular hypothesis)

Unit 4:Polymer Chemistry

Macromolecular hypothesis

Understanding of the nature of polymers

- The growth of polymer industry was restricted by the considerable lack of understanding of the nature of polymer and their unusual properties

- By 1920, the common belief was that Polymer consisted of physically-associated aggregates of small molecules.

- In 1924, Hermann Staudinger proposed the macromolecular hypothesis, which states that polymers are molecules made of covalently bonded elementary units called monomers.

- Staudinger hypothesis was further substantiated by the crystallographic studies of natural polymers reported by Herman Mark and Kurt Meyer.

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Polymers

IUPAC definition:

"A molecule of high relative molecular mass, the structure of which essentially comprises the multiple repetition of units derived, actually or conceptually, from molecules of low relative molecular mass."

Natural Polymers:

- Protein, DNA Starch, Cellulose, polysaccharides, silk, wool, natural rubber
- Textiles for clothing and papyrus

Synthetic Polymers:

- Nylon, Polyethylene, Polyester, Teflon, synthetic rubber
- Light emitting semiconductors, low cost printable devices, as light weight materials for bullet proof vests

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Some Polymers..

Monomers	Polymer	Comments
1. Ethylene $CH_2{=}CH_2$	Polyethylene (PE) $\left[CH_2-CH_2\right]_n$	Moulded objects, tubing, film, electrical insulation, e.g. 'Alkathene', 'Lupolen'.
2. Propylene $CH_2{=}CH(CH_3)$	Polypropylene (PP) $\left[CH_2-CH(CH_3)\right]_n$	Similar uses to PE; lower density, stiffer, e.g. 'Propathene', 'Novolen'.
3. Tetrafluoroethylene $CF_2{=}CF_2$	Polytetrafluoroethylene (PTFE) $\left[CF_2-CF_2\right]_n$	Mouldings, film, coatings; high temperature resistance, chemically inert, excellent electrical insulator, very low coefficient of friction; expensive, e.g. 'Teflon', 'Fluon'.
4. Styrene $CH_2{=}CH$	Polystyrene (PS) $\left[CH_2-CH\right]_n$	Cheap moulded objects, e.g. 'Styron', 'Hostyren'. Modified with rubbers to improve toughness, e.g. high-impact polystyrene (HIPS) and acrylonitrile–butadiene–styrene copolymer (ABS). Expanded by volatilization of a blended blowing agent (e.g. pentane) to produce polystyrene foam.
5. Methyl methacrylate $CH_2{=}C(CH_3)C{=}O(OCH_3)$	Poly(methyl methacrylate) (PMMA) $\left[CH_2-C(CH_3)(C{=}O)(OCH_3)\right]_n$	Transparent sheets and mouldings; used for aeroplane windows; more expensive than PS, e.g. 'Perspex', 'Diakon', 'Lucite', 'Oroglass', 'Plexiglas'.
6. Vinyl chloride $CH_2{=}CH(Cl)$	Poly(vinyl chloride) (PVC) $\left[CH_2-CH(Cl)\right]_n$	Water pipes and gutters, bottles, gramophone records; plasticized to make PVC leathercloth, raincoats, flexible pipe and hose, toys, sheathing on electrical cables, e.g. 'Darvic', 'Welvic', 'Vinoflex', 'Hostalit'.

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Classification of Polymers

- Molecular Architecture
- Chemical Structure
- Origin
- Order of Monomers
- Mechanical properties
- Material properties
- Tacticity
- Nature of the main chain
- Synthetic methods

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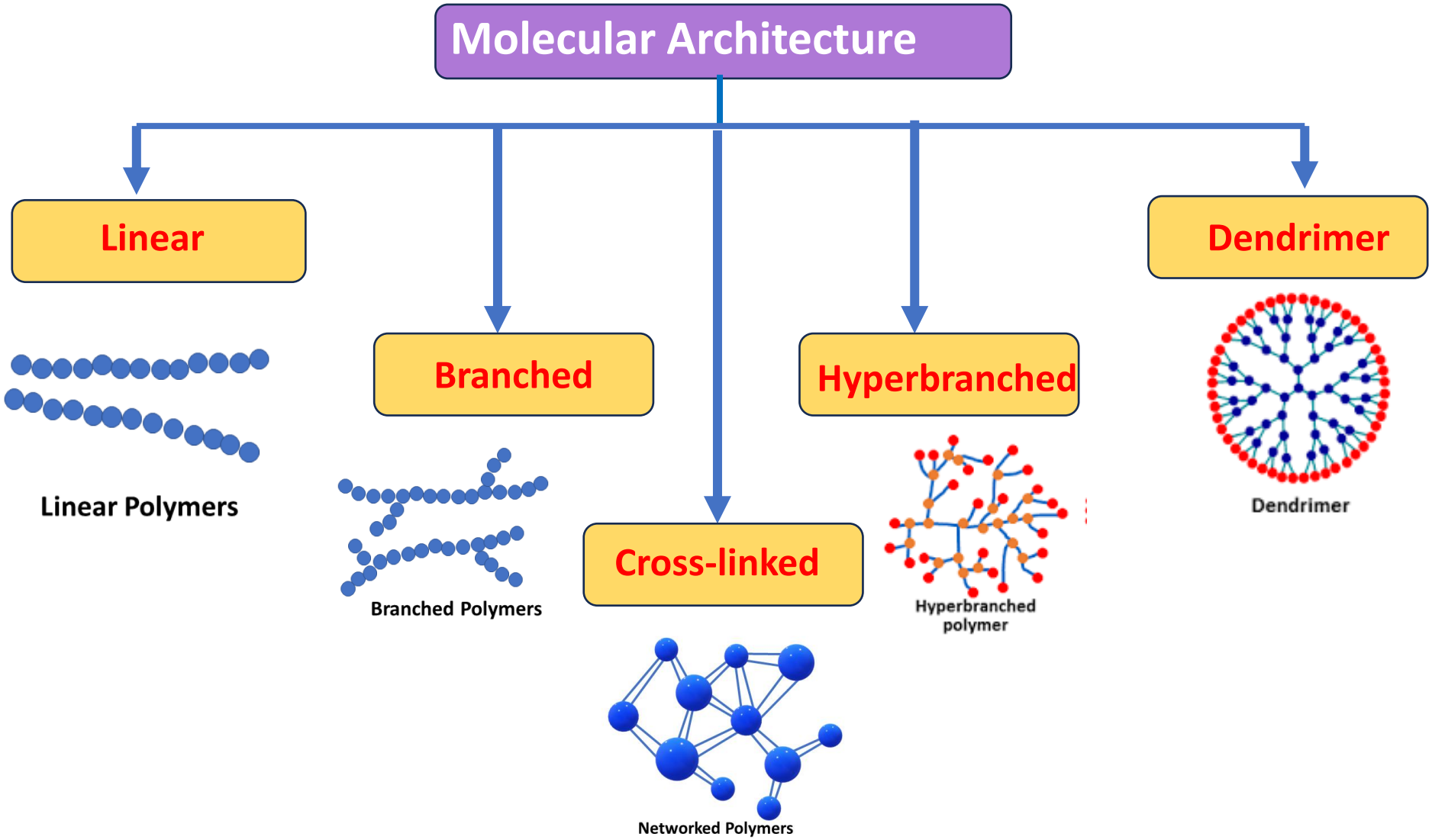
Classification of Polymers



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Classification of Polymers

Chemical Structure

Homopolymer

Polymer derived from one species of monomer

—M_1—M_1—M_1—M_1—

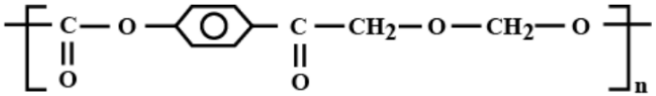
Polyethylene, Polypropylene.

Copolymer

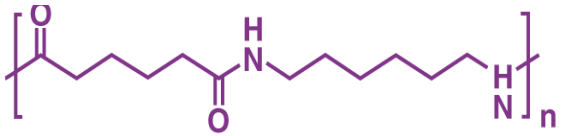
Polymer derived from more than one species of monomer

—M_1—M_2—M_1—M_2—

Nylon, Terylene



Terylene



Structure of Nylon – 6,6

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Classification of Polymers

Order of Monomers

Statistical copolymer

Distribution of monomers obeys statistical laws

Alternating copolymer

Two different types of repeating units arranged alternately

~~A–B–A–B–A–B–A–B–A–B–A–B~~

Random copolymer

The distribution of repeat units is truly random

~~B–B–B–A–B–B–A–B–A–A~~

Block copolymer

Repeat units exist only in long sequences (blocks)

A–A–A–A–A–A–A–A–A–A–B–B–B–B–B–B–B–B–B–B

A–A–A–A–A–A–A–A–B–B–B–B–B–B–B–B–A–A–A–A–A–A–A–A

Graft copolymer

Branches have different chemical structure than the main chain

```
B–B–B–B–B–B
|
B
|
A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A–A
                        |
                        B
                        |
                        B–B–B–B–B–B–B–B
```

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Classification of Polymers

Mechanical properties

Thermoplastics

Polymers that can be softened through heating and can be melted and reshaped repeatedly

Polythene, Polystyrene

Thermoset

Polymers that hardens permanently when heated or chemically treated

Bakelite
Melamine
Formaldehyde

Elastomers

Polymers, that are capable of recovering their original shape after being stretched to great extents

Natural rubber

Thermoplastics

Crystalline | Amorphous

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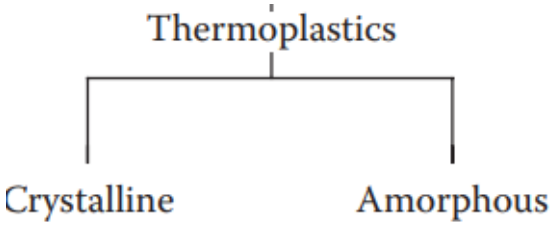
Thermoplastics

Thermoplastics (constitute the largest proportion of the polymers in commercial production)

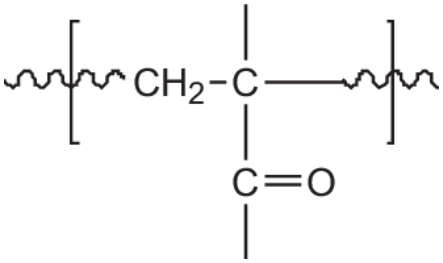
Thermoplastics, often referred to just as plastics, are linear or branched polymers which become liquid upon the application of heat. They can be moulded (and remoulded) into virtually any shape using processing techniques.



- The crystalline phases of such polymers are characterized by their melting temperature Tm, above which such polymers can be converted into artefacts.
- Amorphous polymers are characterized by their glass transition temperature Tg, the temperature at which they transform abruptly from the glassy state (hard) to the rubbery state (soft).

Example:

- Polythene: A well-known thermoplastic used in packaging
- Acrylic: Used as an alternative to glass in items like LCD screens, furniture, vehicle headlights, paint, and lenses



Acrylic

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Elastomers

- Elastomers are crosslinked rubbery polymers that can be stretched easily to high extensions (e.g. 3× to 10× their original dimensions) and which rapidly recover their original dimensions when the applied stress is released.
- This extremely important and useful property is a reflection of their molecular structure in which the network is of low crosslink density.

Examples

- Natural rubber - footwear, clothing, furniture, toys etc.
- Polyurethanes – Lycra clothing, adhesives
- Neoprenes - medical gloves, orthopaedic braces, laptop sleeves

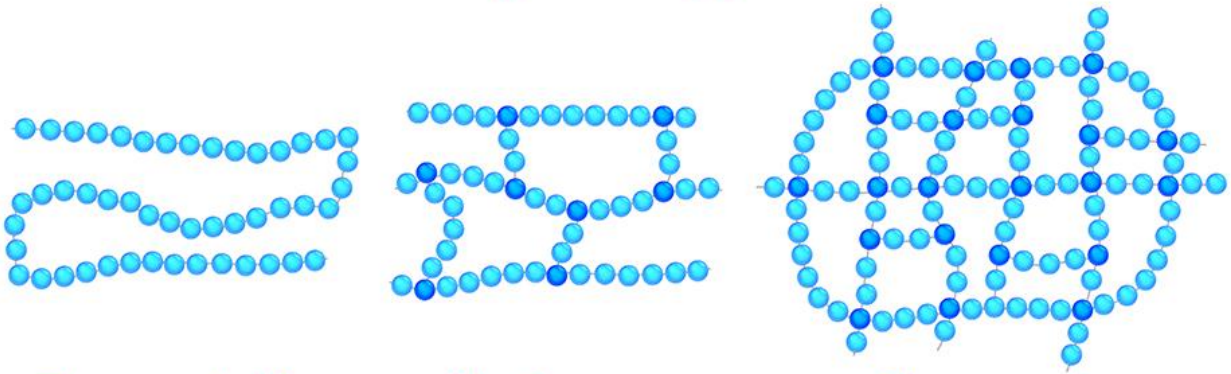
Thermosets

Thermosets normally are rigid materials and are network polymers in which chain motion is greatly restricted by a high degree of crosslinking.

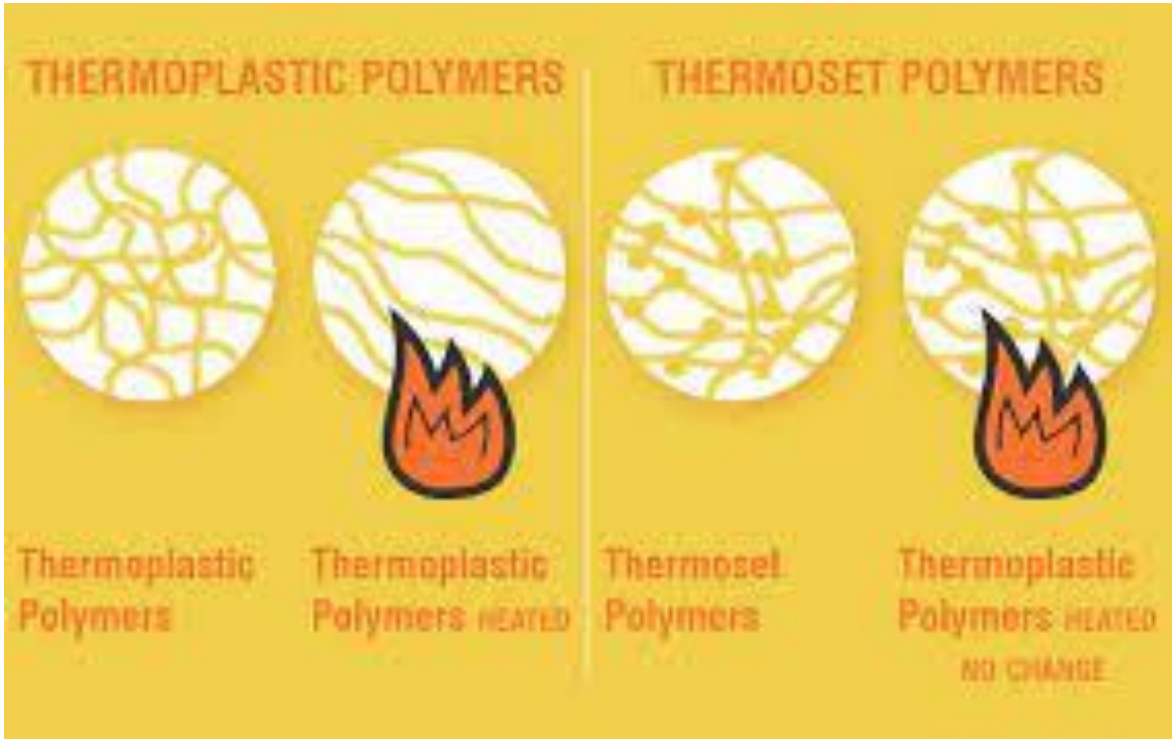
Examples

- Bakelite - electrical switches, saucepan handles

Polymer Types



Thermoplastics Elastomers Thermosets



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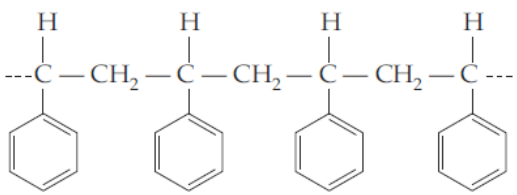
Classification of Polymers

Tacticity

Isotactic

All functional groups are arranged on the same side of the main chain

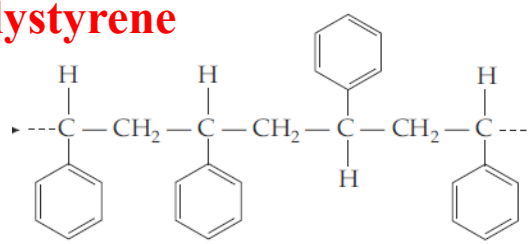


Isotactic polystyrene

Atactic

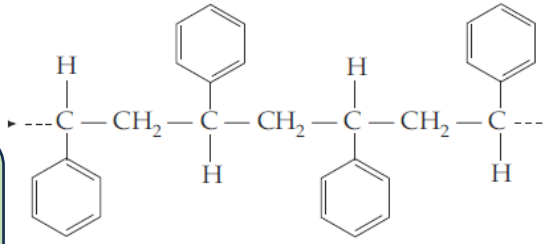
Functional groups are arranged in a random manner around the main chain

Atactic polystyrene



Syn-diotactic

Functional groups are arranged in an alternating fashion around the main chain



Syn-diotactic polystyrene

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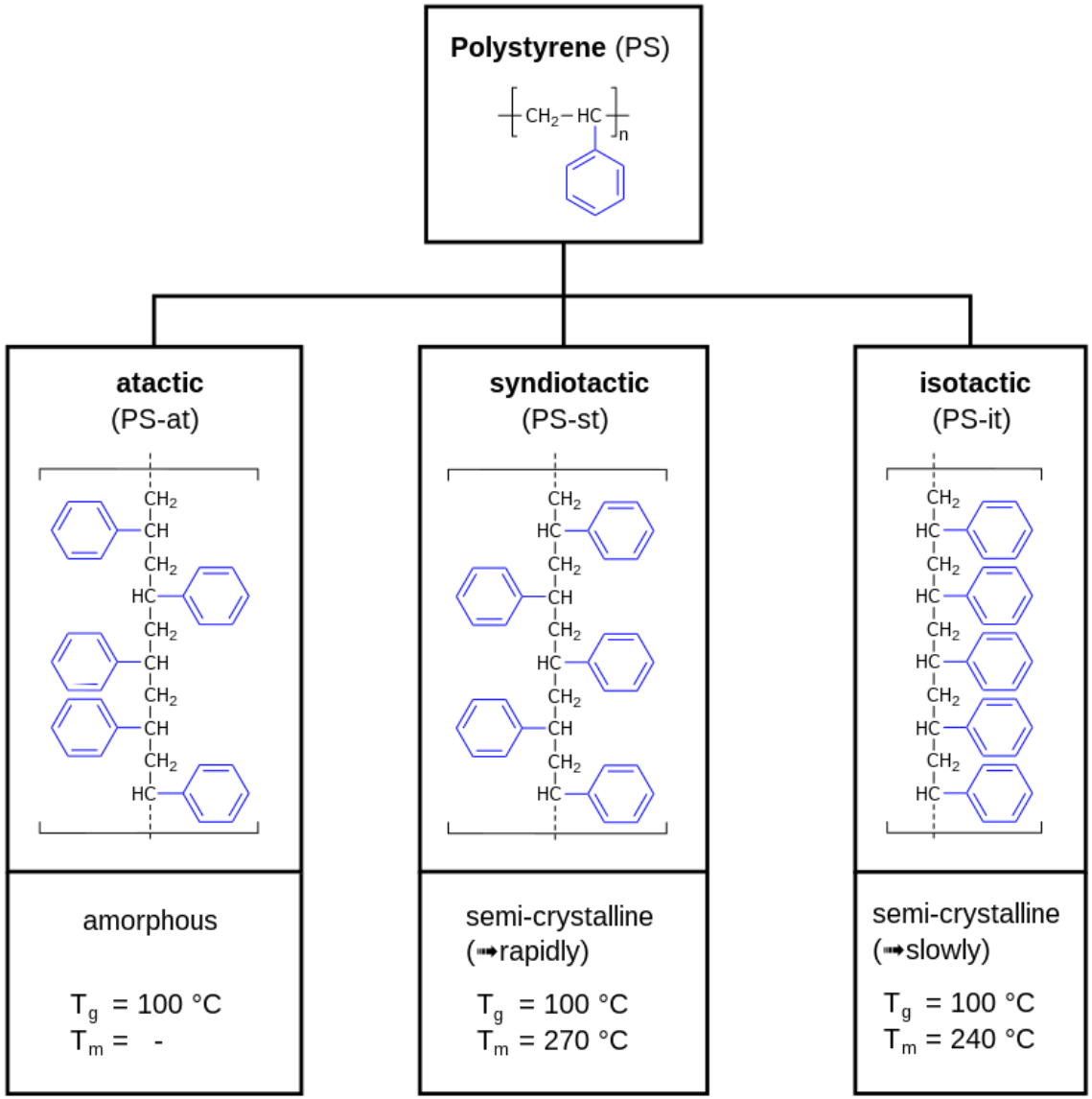
Tacticity

In a polymeric molecule, the arrangement of functional groups, either by an orderly or disorderly manner, with respect to the main chain is called tacticity.

Tacticity affects the physical properties of the polymer.



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Classification of Polymers

Nature of the main chain

Homochain Polymer

The main chain is made up of the same species of atoms

— C — C — C — C — C — C —

Polyethylene, Polyvinyl chloride, Polystyrene, etc.

Heterochain Polymer

The main chain is made up of different atoms

— C — C — N — C — C — N — C — C —

Terylene, Nylon 6:6, Nylon 6, etc.

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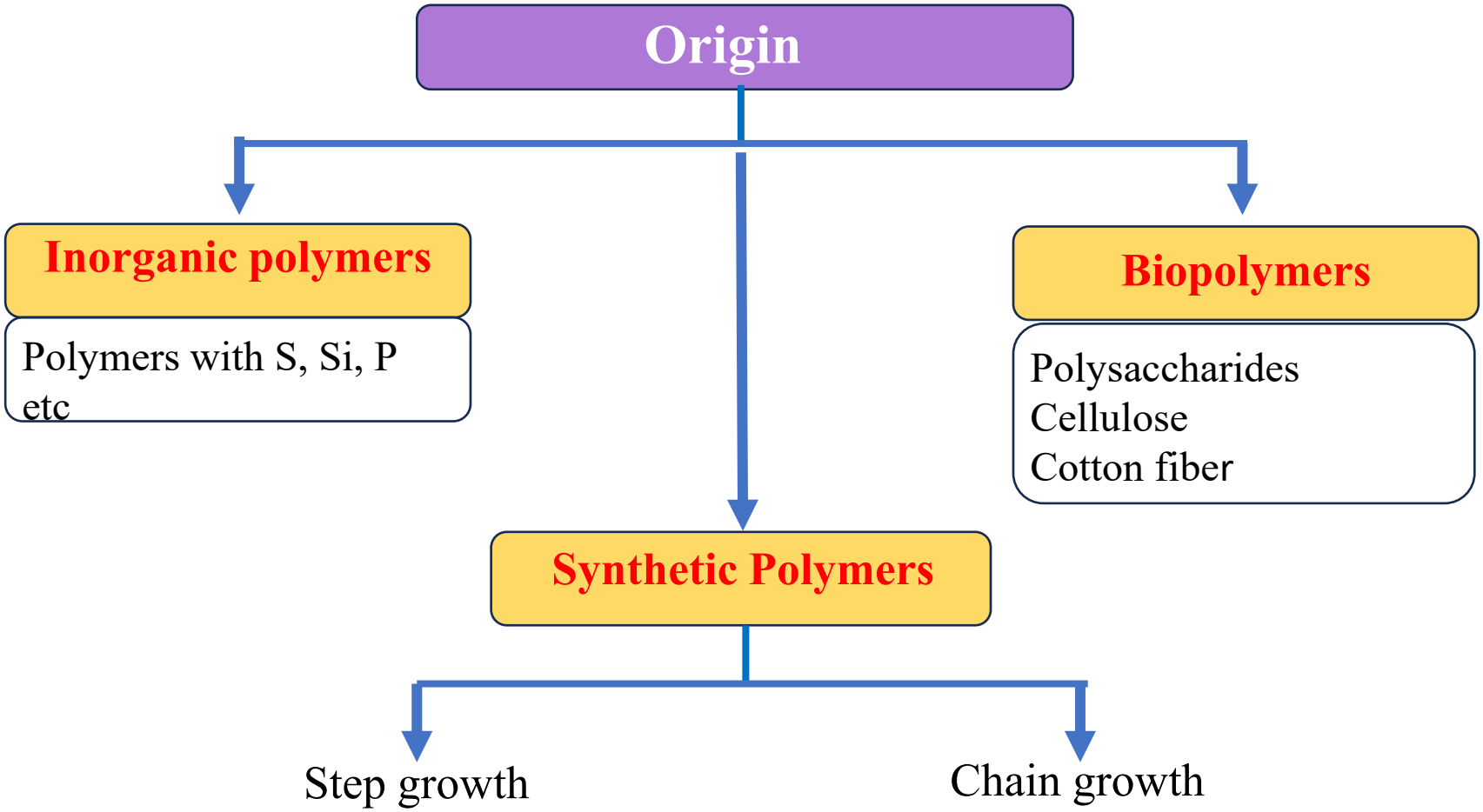
Classification of Polymers



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Nomenclature

The current principles of nomenclature for copolymers are indicated in Table

Type	Connective	Example
Unspecified	-co-	Poly(A-co-B)
Statistical	-stat-	Poly(A-stat-B)
Random	-ran-	Poly(A-ran-B)
Alternating	-alt-	Poly(A-alt-B)
Periodic	-per-	Poly(A-per-B-per-C)
Block	-block-	Poly(A)-block-poly(B)
Graft	-graft-	Poly(A)-graft-poly(B)

A and B represent source- or structure-based names for these repeat units

Examples:

- A statistical copolymer of ethylene and propylene is named ***poly(ethylene-stat-propylene)***
- An AB di-block copolymer of styrene (A) and isoprene (B) is named ***polystyrene-block-polyisoprene***
- An alternating copolymer of styrene and maleic anhydride is named ***poly[styrene-alt-(maleic anhydride)]***

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Molecular weight and distributions

- Many properties of polymers show a strong dependence upon the size of the polymer chains
- A chain form in the course of 1000+ reactions and typically each chain will have different chain lengths
- Molecular weight and molecular weight distributions are essential to characterize a polymer

Molecular weight distribution (MWD) is a measure of the range of molecular weights in a polymer sample. It's a key factor in determining a polymer's physical properties, such as its strength, toughness, and melting point

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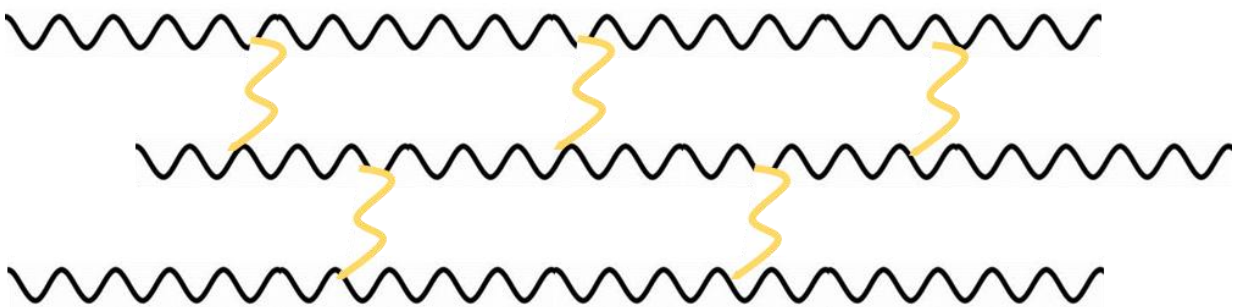
Chain length versus Functional Properties

Example: Polymer based on acrylic acid

Size g/mol	Structure (roughly)	Application
~5000		Dispersant
~70000		Detergent
~2000000		Precipitant
~2000000 Cross linked		Thickeners Super-absorbers

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How do we characterize a molar mass distribution?

Molar mass of a polymer (the mass of 1 mol of the polymer) $M = xM_0$

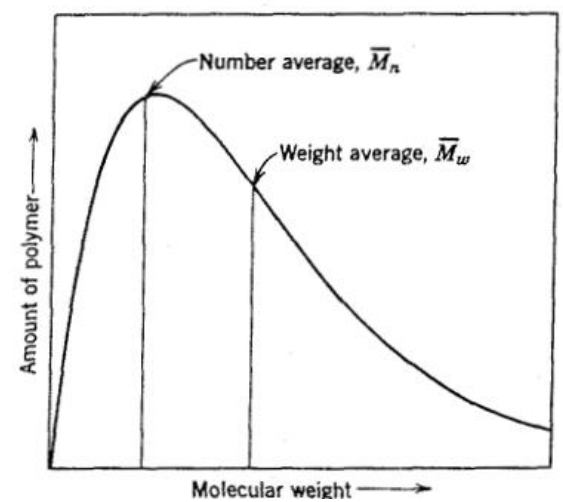
x = degree of polymerization; M_0 = molar mass of the repeat unit

Number-average molar mass $M_n = \sum X_i M_i$ $\quad M_n = \frac{\sum_i N_i M_i}{\sum_i N_i}$

Mass-average molar mass $M_m = \sum w_i M_i$ $\quad M_m = \frac{\sum_i N_i M_i^2}{\sum_i N_i M_i}$

$$X = \text{mole fraction} = \frac{N_i}{\sum N_i}$$

$$w = \text{weight fraction} = \frac{w_i}{\Sigma w_i} = \frac{N_i M_i}{\sum N_i M_i}$$



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How do we characterize a molecular mass distribution?

Number-average molecular mass: $\boldsymbol{M_n}$

- The number average molecular weight (M_n) measuring system requires counting the total number of molecules in a unit mass of polymer irrespective of their shape or size. It means all molecules are treated equally.
- This kind of measuring system is required in cases where certain properties are dependent only upon the number of molecules or repeating units and not upon their weight or sizes.
- These are called colligative properties such as boiling point elevation, freezing point depression, vapor pressure depression, and osmotic pressure changes.

$$M_n = \frac{\sum_i N_i M_i}{\sum_i N_i}$$

M_i is the molecular weight of a chain
N_i is the number of chains of that molecular weight
i is the number of polymer molecules

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How do we characterize a molecular weight distribution?

Mass-average molecular mass: M_m

- Mass average molecular mass (M_m) measuring system includes the mass of individual chains, which contributes to the overall molecular mass of the polymer. The mass average molecular mass favors large molecules
- It is based on the fact that bigger molecules contain more mass than smaller molecules. Thus larger the size of molecule makes more contribution to the molecular weight average.
- Unlike M_n, M_m accounts for the molecular size instead of just the numbers. A classic example of property that depends on weight average molecular weight is light scattering.

$$M_m = \frac{\sum_i N_i M_i^2}{\sum_i N_i M_i}$$

M_i is the molecular mass of a chain
N_i is the number of chains of that molecular mass
i is the number of polymer molecules

Unit 4:Polymer Chemistry

How do we characterize a molecular weight distribution?

Number-average molecular mass:M_n

N_1 molecules have molecular mass M_1 each
N_2 molecules have molecular mass M_2 each
N_3 molecules have molecular mass M_3 each

Total molecular mass of all the N_1 molecule = N_1M_1
Total molecular mass of all the N_2 molecule = N_2M_2
Total molecular mass of all the N_3 molecule = N_3M_3

The sum of all molecular masses
$N_1M_1 + N_2M_2 + N_3M_3 = \Sigma N_iM_i$

Total number of molecules = $N_1 + N_2 + N_3 = \Sigma N_i$

Hence the number-average molecular mass, M_n is given by

$$M_n = \frac{\sum_i N_i M_i}{\sum_i N_i}$$

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How do we characterize a molecular weight distribution?

Mass-average molecular mass: M_m

N_1 molecules have molecular mass M_1 each
N_2 molecules have molecular mass M_2 each
N_3 molecules have molecular mass M_3 each

Total molecular mass of all the N_1 molecule = N_1M_1
Total molecular mass of all the N_2 molecule = N_2M_2
Total molecular mass of all the N_3 molecule = N_3M_3

The products with their respective molecular masses will be
$N_1M_1 * M_1 + N_2M_2 * M_2 + N_3M_3 * M_3 = \Sigma N_iM_i^2$

Total mass of all the molecules = $N_1M_1 + N_2M_2 + N_3M_3 = \Sigma N_iM_i$

Hence the mass-average molecular mass, M_n is given by

$$M_m = \frac{\sum_i N_i M_i^2}{\sum_i N_i M_i}$$

Unit 4:Polymer Chemistry

Condition under which $M_n = M_m$

$$M_n = \frac{N_1M_1 + N_2M_2 + N_3M_3}{N_1 + N_2 + N_3}$$

$$M_m = \frac{N_1M_1^2 + N_2M_2^2 + N_3M_3^2}{N_1M_1 + N_2M_2 + N_3M_3}$$

If $M_n = M_m$

$$\frac{N_1M_1 + N_2M_2 + N_3M_3}{N_1 + N_2 + N_3} = \frac{N_1M_1^2 + N_2M_2^2 + N_3M_3^2}{N_1M_1 + N_2M_2 + N_3M_3}$$

The two sides can be equal only if $M_1 = M_2 = M_3$

When all the molecules have the same molecular mass, the average has no significance

Unit 4:Polymer Chemistry

Problem 1:

A Suspension contains equal masses of particles with molecular masses 20000 and 40000. Calculate M_n and M_m

equal masses of particles $N_1 M_1 = N_2 M_2$

$$M_n = \frac{N_1 M_1 + N_2 M_2}{N_1 + N_2}$$

$$M_m = \frac{N_1 M_1^2 + N_2 M_2^2}{N_1 M_1 + N_2 M_2}$$

$$M_n = 26667$$

$$M_m = 30000$$

Problem 2:

Calculate M_m for a system containing equal number of particles with molecular masses 10000 and 20000